6.3 The Evolution Of Graphics Cards

The video card, per se, made its appearance a long time ago, even before the term was coined. It was only with the introduction of 3D accelerators that users really got up and took notice of what video cards were able to do to visuals. Graphics and games never remained the same again with the introduction of the 3D graphics accelerator.

Video cards were referred to as video adapters, and were simply used to generate text and primitive computer graphics on the monitor (or Video Display Unit as it was called a long time ago). Video support was extremely limited, and was in the form of sprites i.e., animated images that form objects or characters in 2D games. Simply put, they are bitmaps of arbitrary shapes that can be moved across complex backgrounds. This was because the video adapters by themselves were processor-limited, and the system processor in most cases was without a floating-point unit processor—which restricted much of the computation required to render graphics onscreen.

The XFX GeForce 7800 GTX 256MB

With the evolution of and innovations in chip technology, microprocessors moved from the realm of being just system processors to being graphics processing units. After all, any general purpose CPU can be transformed to be a dedicated graphics processing unit. However, these cards with microprocessors were expensive, and for video cards to hit the desktop PC segment, vendors thought of an innovative way.

With the introduction of Microsoft Windows, rendering graphics on the PC took a new course. At that same time, vendors were able to make scaled down versions of microprocessor-based dedicated graphics cards, called 2D accelerators. These cards had drawing support and also had a frame buffer, which could cache